Machine Learning Methods for Finance

Unsupervised Learning - Clustering

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Outline

1. Clustering
   - K-Means
   - Gaussian Mixture Modeling
   - Hierarchical Clustering
   - Minimum Spanning Tree

Motivation

- These notes accompany the Duke Economics course *Machine Learning for Finance*.
- The course focuses on how machine learning methods can be used to design, evaluate, and implement systematic investment strategies.
- Our goal is to introduce a range of ML techniques with enough mathematical structure to build intuition, while avoiding unnecessary formalism.
- Throughout the notes we use a simple return–momentum signal as a running example. Although momentum is not always the ideal illustration for every technique, using a single, familiar signal provides a consistent thread. This is especially true when we transition to real data and empirical implementation.
- The methods we discuss are broadly applicable across many signals, asset classes, and investment settings.

Description of ML Methods

Unsupervised Learning

- **Clustering**
  Group similar assets by momentum characteristics
- **Dimension Reduction**
  Condense multiple momentum measures into fewer factors
- **Generative Modeling**
  Simulate returns to derive momentum structure

Supervised Learning

- **Classification**
  Label assets as high/low momentum
- **Regression**
  Predict cross-sectional returns using momentum inputs

Time Series Modeling

- **Unsupervised**
  Group assets by temporal behavior, then measure momentum
- **Supervised**
  Forecast returns based on persistence

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Clustering Momentum

Q: How form a momentum signal via Clustering?

1. Convert prices to returns, then compute momentum
   - Jegadeesh and Titman:
   - Technical Analysis style:
2. Group assets by

$$s_{i,t} = \mathbf{1}\left[p^{(+)}(X_{i,t}, Z_t)\right] - \mathbf{1}\left[p^{(-)}(X_{i,t}, Z_t)\right]$$

$$p^{(+)} = \mathbf{1}\left[k_t^i = k_t^{(+)}\right] \quad p^{(-)} = \mathbf{1}\left[k_t^i = k_t^{(-)}\right]$$

where

-
- $X_{it} =$
- $\bar{m}_{k,t} = \frac{1}{N_{k,t}} \sum_{i:\, k_t^i = k} mom_{i,t}$
- $mom_{i,t}$: momentum of asset i
- k_t^i:
- $\bar{m}_{k,t}$:

Types of Clustering

- Hard:
- Soft:

Clustering Methods

1. Centroid-based —
2. Model-based —
3. Hierarchical —
4. Graph-based —
5. Density-based —

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K-Means

The most common Centroid Based, Hard Clustering method

1. Choose K as the number of clusters
2. Assign each observation x_i to one of the K clusters
3. Compute the mean (aka centroid) of each cluster $\bar{x}^k$
4. For each observation
   - Update cluster assignments by
   - Update cluster means by
   - Repeat until

Output

-
-
-

Considerations for Implementing K-Means

1. Accommodating multidimensional x
2. Choosing the number of clusters
3. Initializing cluster assignments
4. Pre-processing — helpful to standardize, or otherwise scale, the data before clustering, especially when using Euclidean distance
5. Tie Breaking
6. Empty Clusters
7. Stopping Criterion

Note: We won't dwell on 5,6,7 in the list above.

Multidimensional x

Suppose $\mathbf{x}_i^k$ is not a scalar; e.g. Clustering momentum AND volume across the SP500, in which case $dim(\mathbf{x}_i) = 2 \times 1$

- Cluster centroids are columnwise; i.e.
- Clustering Error expands

Ticker	Momentum	Errors	Squared Errors
A	1	-1.5	2.25
B	2	-0.5	0.25
C	3	0.5	0.25
D	4	1.5	2.25
Avg	2.5		1.25
Sum			5

Ticker	Momentum	Volume	Error		Squared Errors		
			Momentum	Volume	Momentum	Volume	Squared Errors
A	1	200	-1.5	-200	2.25	40000	40002.25
B	2	300	-0.5	-100	0.25	10000	10000.25
C	3	700	0.5	300	0.25	90000	90000.25
D	4	400	1.5	0	2.25	0	2.25
Avg	2.5	400					35001.25

Ticker	Volume	Errors	Squared Errors
A	200	-200	40000
B	300	-100	10000
C	700	300	90000
D	400	0	0
Avg	400		35000
Sum			140000

Sum of Avg	35001.25
(1/4)*Sum of Sums	35001.25

Note: Euclidean distance is not the only distance measure available for K-Means

Choosing the number of clusters

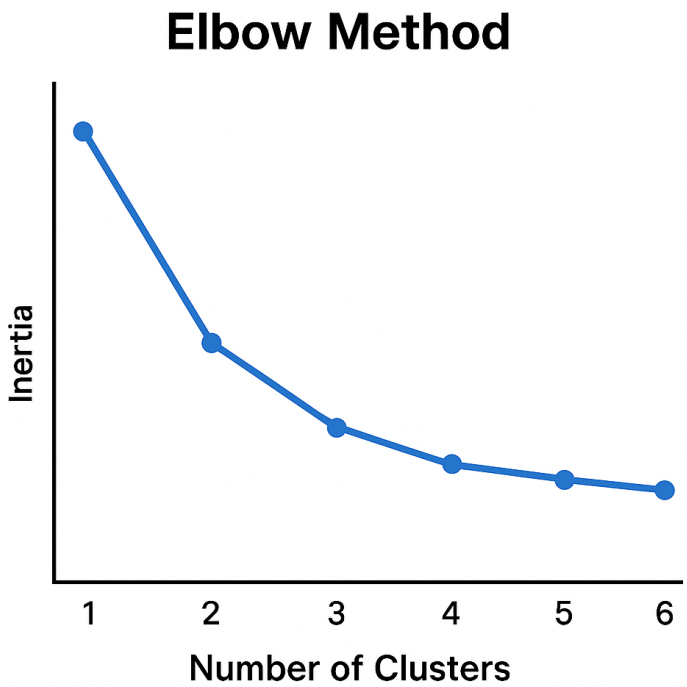
- If we set $K = 1$, then
- If we set $K = N$, then

Typical Approaches

1. Choose K based on context; e.g.
2. If clustering is an aspect of our feature engineering to a supervised ML exercise, we can chose the K that
3. If clustering is primarily part of EDA, then we can

Choosing Number of Clusters with Elbow Method

Q: How many clusters are suggested?



A:

Initializing Cluster Assignments

A few of the many approaches available

-
- (K-means ++)
  1. Pick first centroid randomly
  2. Compute squared distances of each observation to that centroid
  3. Choose next centroid with probability proportional to the squared distance so that points farther away are more likely to be chosen
  4. Repeat until K centroids are chosen.

Judging Cluster Assignment with Silhouette Plots

Silhouette Score

$$s_i = \frac{b_i - a_i}{max\{a_i, b_i\}} \in \{-1, \cdots, +1\}$$

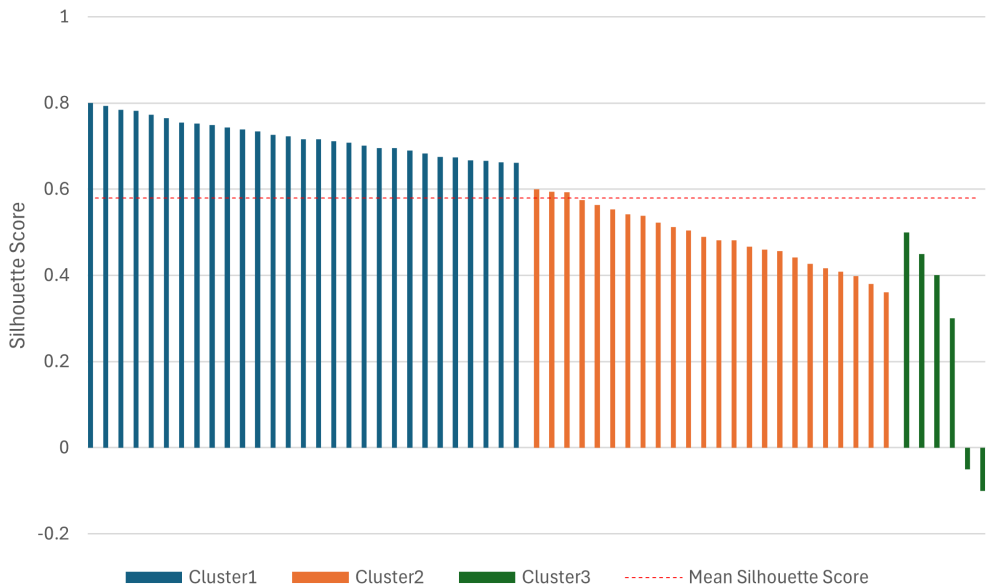
where

- a_i : cluster cohesion;
- b_i : cluster separation;

We compute the mean Silhouette score $\bar{s}$ and compare the individual scores to that mean. If the scores within a cluster are less than the $\bar{s}$, it implies that

Good and Bad Clusters

Q: Which cluster(s) are good/bad? What does a negative score mean?



A:

- Cluster 3 seems
- The negative scores imply

K-Means Clustering Momentum

Q: How form a momentum signal via K-Means Clustering?

1.
2.

$$s_{i,t} = \mathbf{1}\left[p^{(+)}(X_{i,t}, Z_t)\right] - \mathbf{1}\left[p^{(-)}(X_{i,t}, Z_t)\right]$$

$$p^{(+)} = \mathbf{1}\left[k_t^i = k_t^{(+)}\right] \quad p^{(-)} = \mathbf{1}\left[k_t^i = k_t^{(-)}\right]$$

where

- $k_t^{(+)} = \arg\max_k \bar{m}_{k,t}$; $k_t^{(-)} = \arg\min_k \bar{m}_{k,t}$
- X_{it} = momentum for asset i; Z_t = momentum for other assets
- $\bar{m}_{k,t} = \frac{1}{N_{k,t}} \sum_{i:\, k_t^i = k} mom_{i,t}$
- $mom_{i,t}$: momentum of asset i
- k_t^i: cluster label
- $\bar{m}_{k,t}$: average cluster-level momentum

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Motivating Gaussian Mixture Models for Momentum

Common application of Centroid Based, Soft Clustering

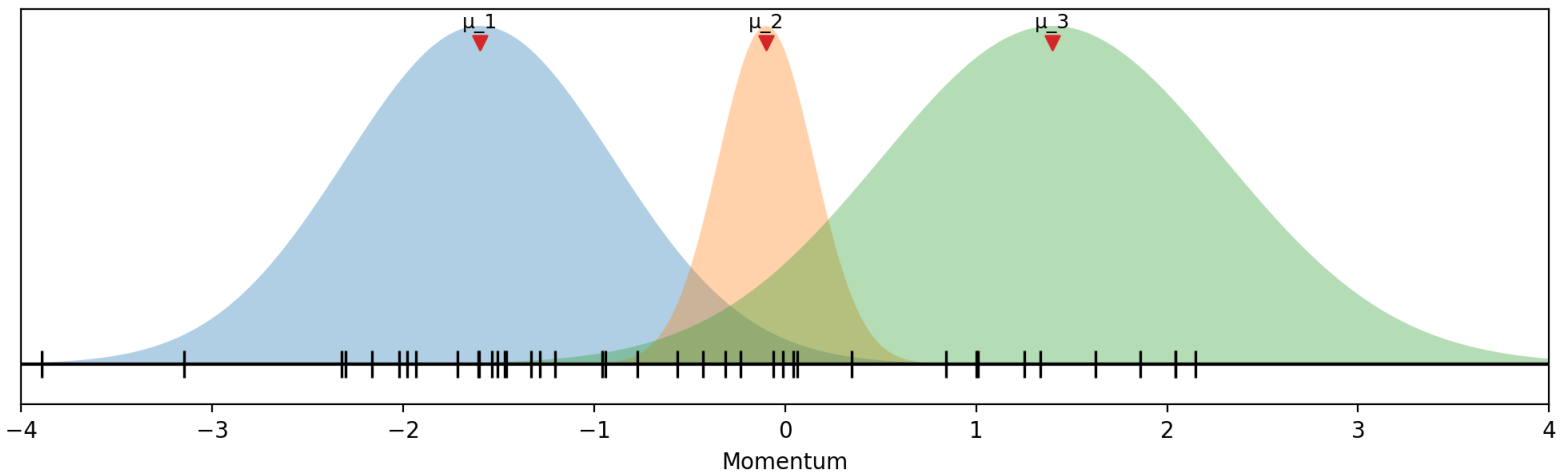
- Consider Momentum as a *continuous*, *noisy* signal
- There are no natural hard cutoffs between:
  - Losers
  - Neutral assets
  - Winners
- Assets near the boundary may plausibly belong to multiple regimes

GMM perspective:

- Observed momentum is
- Each regime has its own
- Each asset has a

Motivating Gaussian Mixture Models (GMM)

Common application of Centroid Based, Soft Clustering



Steps for GMM Clustering

1. Initialize number of clusters K, cluster means $\bar{x}^k$, covariances Σ_x^k, and prior probabilities p^k that i is a member of cluster k
2. Update "degrees of belonging" (posterior probability that observation i belonging to cluster k) as $\hat{p}(k^i = k|x) = \frac{\hat{p}^k N(x_i; \hat{\bar{x}}^k, \Sigma_x^k)}{\sum_{k'=1}^K \hat{p}^{k'} N(x_i; \hat{\bar{x}}^{k'}, \Sigma_x^{k'})}$
3. For each cluster k update
   - Cluster probability: $\hat{p}^k = (N_k/N)$ with $N_k = \sum_{i=1}^N \hat{p}(k^i = k|x)$
   - Cluster mean: $\hat{\bar{x}}^k = (1/N_k) \sum_{i=1}^N \hat{p}(k^i = k|x) \times \mathbf{x}_i$
   - Cluster covar: $\hat{\Sigma}_x^k = (1/N_k) \sum_{i=1}^N \hat{p}(k^i = k|x) \times (\mathbf{x}_i - \hat{\bar{x}}^k)(\mathbf{x}_i - \hat{\bar{x}}^k)'$
   - Repeat until convergence

Output: predicted "degrees of belonging" $\hat{p}(k^i = k|x)$; i.e. probability that asset i is a member of each of the K clusters

GMM Clustering Momentum

Q: How form a momentum signal via GMM Clustering?

1. Choose the number of clusters K via
2. Group assets by momentum via

$$s_{i,t} = \mathbf{1}\left[p^{(+)}(X_{i,t}, Z_t)\right] - \mathbf{1}\left[p^{(-)}(X_{i,t}, Z_t)\right]$$
$$p^{(+)} = \mathbf{1}\left[k_t^i = k_t^{(+)}\right] \quad p^{(-)} = \mathbf{1}\left[k_t^i = k_t^{(-)}\right]$$

where

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Hierarchical Clustering

Hierarchical clustering groups data within a tree structure based on similarity

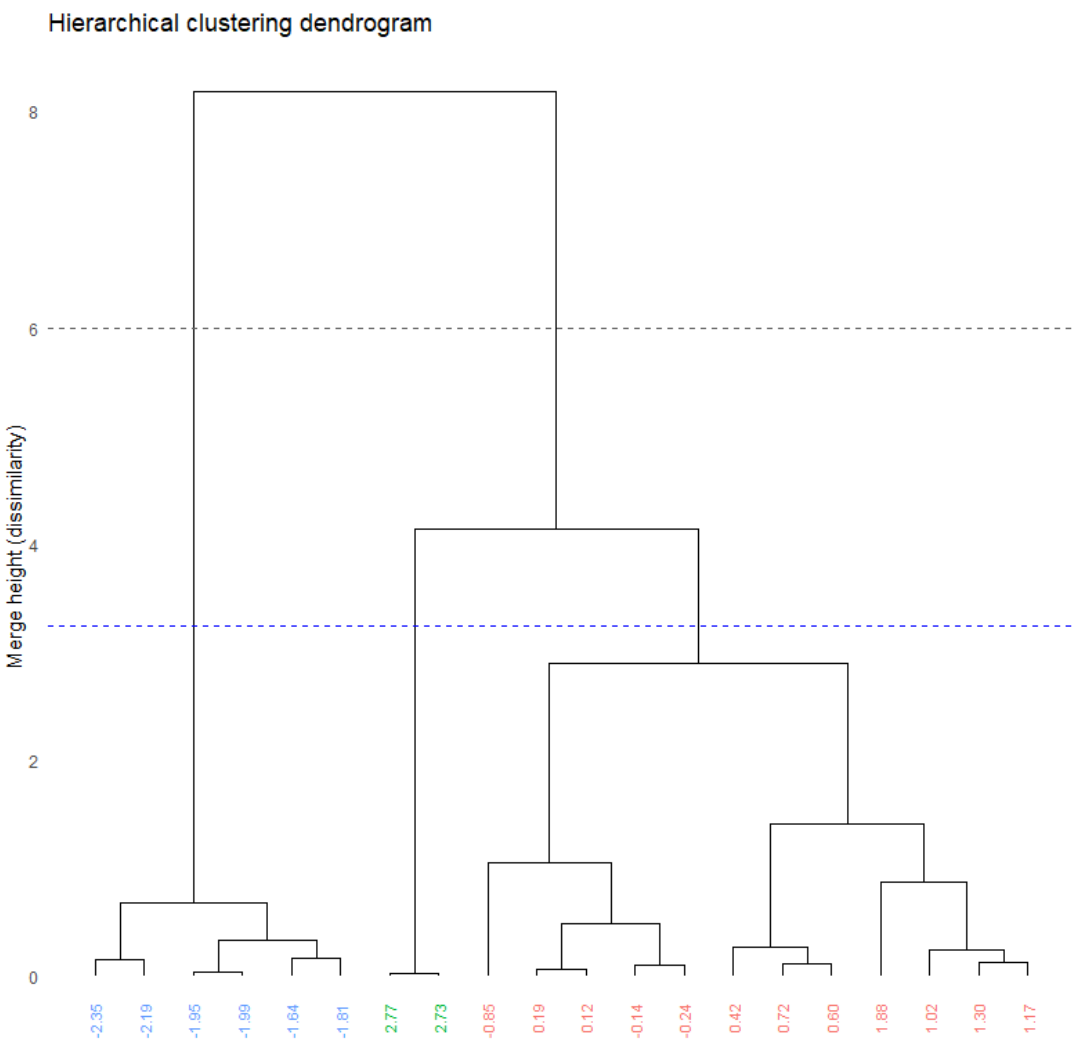
- Consider Momentum as a
- Assets may not be
- They are attractive

Hierarchical Clustering perspective:

- Arrange assets by
- Produce a
- Choose the number of clusters

Hierarchical Clustering - Dendrogram

Suppose the horizontal axis measures momentum, and each observation is another asset



Q: Why is the vertical line for 2.77 and 2.73 so large?

A:

Q: How many clusters chosen with the blue horizontal line?

A:

Q: How many clusters chosen with the gray horizontal line?

A:

Ways to create the hierarchy

- Agglomerative: Start with each observation as a cluster and merge the closest pairs until a single cluster remains (typical)
- Divisive: Start with a single cluster and iteratively split into smaller clusters until each cluster contains only one observation

Steps for Agglomerative Clustering

1. Compute a pair-wise distance matrix of all observations e.g. $d(i,j) = (x_i - x_j)^2$
2. Initialize with one cluster per observation e.g. N clusters; one for each asset
3. Merge the two closest clusters based on either
   - Single linkage: distance = min pairwise distance between members
   - Complete linkage: distance = max pairwise distance between members
   - Average linkage: distance = average pairwise distance
   - Ward's method: merge clusters that minimize total within-cluster variance
4. Update distance matrix to reflect the new cluster
5. Repeat until one cluster remains

Adapting Hierarchical Clustering for Multidimensional x

Suppose we are clustering on 12mth and 2mth returns such that $x_i = (12mth, 2mth)$

1. Compute a pair-wise distance matrix of all observations e.g., $d(i,j) = (1/2)[(12mth_i - 12mth_j)^2 + (2mth_i - 2mth_j)^2]$
2. Initialize with one cluster per observation e.g. N clusters; one for each asset
3. Merge the
4. Update
5. Repeat until

Hierarchical Clustering Momentum

Q: How form a momentum signal via Hierarchical Clustering?

1.
2.

$$s_{i,t} = \mathbf{1}\left[p^{(+)}(X_{i,t}, Z_t)\right] - \mathbf{1}\left[p^{(-)}(X_{i,t}, Z_t)\right]$$

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Minimum Spanning Tree (MST)

MST connects all observations using only pairwise distances, choosing edges so that the total connection cost is minimized

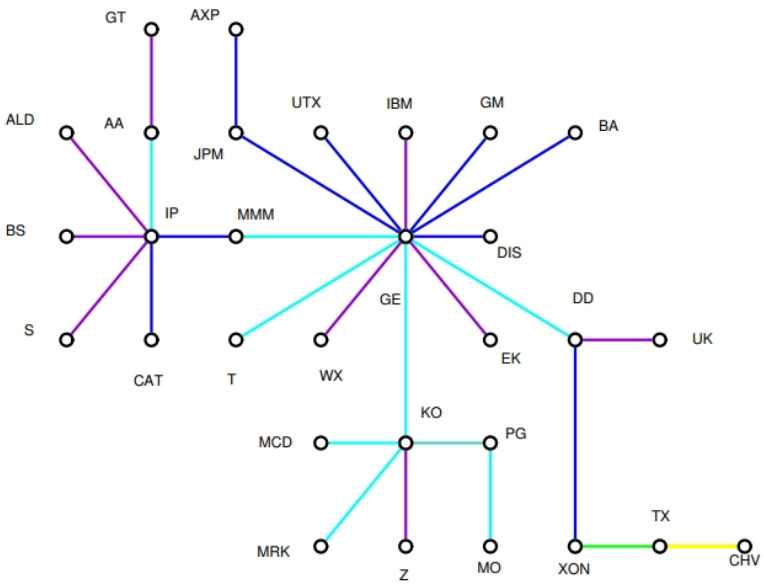


Figure: Mantegna, R. N. *Hierarchical Structure in Financial Markets*. arXiv:cond-mat/9802256 (1998)

Building an MST from Return Correlations

1. Compute asset returns (e.g., daily log returns)
2. Compute the return correlation matrix
3. Convert correlations to a distance metric:

$$d_{ij} = \sqrt{2(1 - \rho_{ij})} \in [0, 2]$$

4. Construct the Minimum Spanning Tree (MST) from the distance matrix
    1. Initialize with any asset (e.g. Prim's algorithm)
    2. Iteratively add the asset whose distance to the existing tree is smallest (i.e., most highly correlated with at least one asset already in the tree)
    3. Continue until all assets are connected by $N - 1$ edges

Measuring Centrality

Centrality quantifies how much a network is concentrated around a single asset

Here are a few common centrality measures for asset (i):

1. Degree Centrality: # of edges (Number of connections)
   - Q: Why shouldn't we use this centrality measure for asset returns?
   - A:
2. Weighted Degree Centrality: $\sum d_{ij}$ (Strength of connections)
3. Eigenvector Centrality: (Strength of connections to other important assets)
   - Define distance matrix D whose elements are d_{ij}, eigenvector c and eigenvalue λ
   - Find combo's of $\{c, \lambda\}$ such that $Dc = \lambda c$
   - Normalize $\sum c_i = 1$
   - Eigenvector Centrality: $c_i = \frac{1}{\lambda} \sum_{i=1}^{N} W_{ij} c_j$ (higher is more central)

Clustering With MST

1. Form the MST, which produces
2. Remove the
3. The remaining graph is a forest;

MST Clustering Momentum

Q: How form a momentum signal via MST Clustering?

1.
2.

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